

Activity trace — extending matched SHD training

ar071 · 19 July 2026 · Draft

Checkpoint CP-001

This publication-to-mandate checkpoint has immutable private source SHA-256 prefix `4b08b55d459f`. Its checkpoint time is `2026-07-19 00:49:00.457 UTC`. It has no predecessor within this night. Hidden reasoning, tool payloads, injected environment material, credentials, private paths, and infrastructure identifiers are excluded; the one visible private local link is explicitly redacted below.

Decision and action ledger

1. Pull request 51 was explicitly accepted and merged into `main` using merge commit `569926f`; no squash or rebase was used.
2. The accepted result motivates higher accuracy, but repeated official-test use was rejected as an optimization loop. The next outcome is held-out validation accuracy on `exp068`'s identical deterministic split.
3. The approved intervention changes only fixed training duration, from 40 to 80 epochs, in both matched cells. One seed remains appropriate for this exploratory mechanism check.
4. New identifiers are `ar070`, `ar071`, and `exp069`. At approval, `exp069` spend was 0 USD, no pod was active, and no implementation or result existed.

Privacy-sanitized visible transcript

Assistant

2026-07-19 00:41:04.944 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

`Exp068` is complete and stopped at the human-review gate. `PR 51` remains draft and unmerged.

Key evidence:

- PING: **45.72%** official-test accuracy
- COBA: **35.07%**
- Difference: **+10.64 percentage points**
- PING/COBA macro accuracy: 45.47% / 34.64%
- PING/COBA cross-entropy: 2.055 / 2.345
- PING E/I activity: 10.50 / 47.99 Hz
- COBA E activity: 28.44 Hz
- One seed; all 2,264 official-test samples evaluated after paired checkpoint freeze
- Zero skipped updates or non-finite batches
- Exact recorded compute spend: **2.4414148 USD**
- Active pods: **0**

Links included PR 51 and the rendered ar068, exp068, and ar069 pages. (PRIVATE LOCAL PATH REDACTED)

Validation passed: focused Ruff and ty, compilation, 20 artifact hashes, registered-result assertions, raster inspection, zero-stub 35-entry Demolab build, and GitHub preview. Repository-wide CI lint remains red solely from previously documented ty debt; the test job was still running at handoff.

Final commits: 7671ce4 and ae7a26a. Goal usage: 5,630 tokens over roughly 18 seconds of tracked goal execution.

User

2026-07-19 00:44:59.911 UTC

Role: user · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

merge into main, then make a new pr we gonna try get those acc numbers higher

Assistant

2026-07-19 00:45:05.441 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

I'll merge PR 51 into `main` at the publication gate first, preserving its merge history. Then I'll synchronize `main` and open a fresh research branch/PR for the accuracy-improvement night; I'll keep that new experiment at the mandate-design gate until its scientific setup is explicit. I'm using the GitHub workflow guidance because this changes PR and branch state.

Assistant

2026-07-19 00:45:15.365 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR 51 is mergeable and the preview is green. Its lint failure is still only the previously documented repository-wide type debt, with no exp068 diagnostic; the long test job remains in progress. Your explicit merge decision resolves the publication gate, so I'm marking the draft ready and merging it now without squashing.

Assistant

2026-07-19 00:45:28.680 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR 51 is merged into `main` as merge commit 569926f. This closes the exp068 publication gate. I'm synchronizing that accepted state and checking the autoresearch registration rules and next available IDs before creating the new accuracy-improvement PR.

Assistant

2026-07-19 00:46:05.090 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

PR 51 is merged into `main` as merge commit `569926f`.

Before opening the next PR, the autoresearch mandate gate requires a locked scientific design. I recommend this clean next step:

- One seed, same deterministic 7,340/816 SHD development split.
- Keep the official test sealed; repeated use would turn it into an optimization target.
- Train matched COBA and PING for 80 epochs instead of 40.
- Preserve every `exp068` architecture, optimizer, input, and readout setting.
- Select checkpoints using validation accuracy, then cross-entropy, then earlier epoch.
- Primary criterion: PING improves at least 3 percentage points over its `exp068` 43.50% validation baseline.
- Secondary: whether COBA improves over 40.81%, whether PING remains ahead, and activity/runtime diagnostics.
- One local smoke, then two concurrent pods, 10 USD ceiling, no `tools/snn` edits.
- New IDs: `ar070`, `ar071`, and `exp069`.
- Only after approval: lock the mandate on `main`, create the night branch, and open the new draft PR.

Reply **go** to approve this mandate.

User

2026-07-19 00:48:51.919 UTC

Role: user · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

go

Assistant

2026-07-19 00:49:00.457 UTC

Role: assistant · Checkpoint: CP-001 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

Mandate approved. I'm locking `ar070`, `ar071`, and `exp069` on `main` now with the 80-epoch, validation-only exploratory contract. This approval starts no compute: after the anchor is pushed, I'll create the dedicated night branch and new draft PR from that exact commit.

Checkpoint CP-002

This branch-and-PR checkpoint follows CP-001. Its immutable private source has SHA-256 prefix `77b86b9fea8d`, and its checkpoint time is `2026-07-19 00:53:19.217 UTC`.

Action ledger

1. The approved mandate was committed and pushed to `main` at `3c440aa`.
2. The night branch was created from that exact anchor and its first Record checkpoint was committed at `8229b83`.
3. Draft pull request 52 is open. No experiment runner, smoke, pod, result, or cost exists; `exp069` spend remains 0 USD.

Visible messages added in CP-002

Assistant

2026-07-19 00:50:48.822 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The mandate itself builds, but the initial activity-trace rendering found an unclosed Typst delimiter in a verbatim message containing bracket and currency punctuation. I'm escaping only that transcript presentation; the approved scientific contract remains unchanged.

Assistant

2026-07-19 00:52:07.816 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The mandate is now preregistered on `main` at `3c440aa`; `exp069` spend is 0 USD and no compute has started. I'm creating `night/spiking-heidelberg-digits/ar070` from that exact anchor, adding the branch-start checkpoint, then pushing and opening the new draft PR.

Assistant

2026-07-19 00:53:19.217 UTC

Role: assistant · Checkpoint: CP-002 · Session: 019f76ab-932b-72c2-b163-463ead65d8d5

The new draft PR is open as 52 from the exact locked anchor. I'm adding its timestamped branch/PR checkpoint to `ar071` and pushing that final setup commit; `exp069` remains unrun with 0 USD spend and no active pods.